Digital Twin for Epileptor Seizure Transition Prediction

Statistically Conditioned Digital Twin (SCDT) from Normal Dynamics Alone

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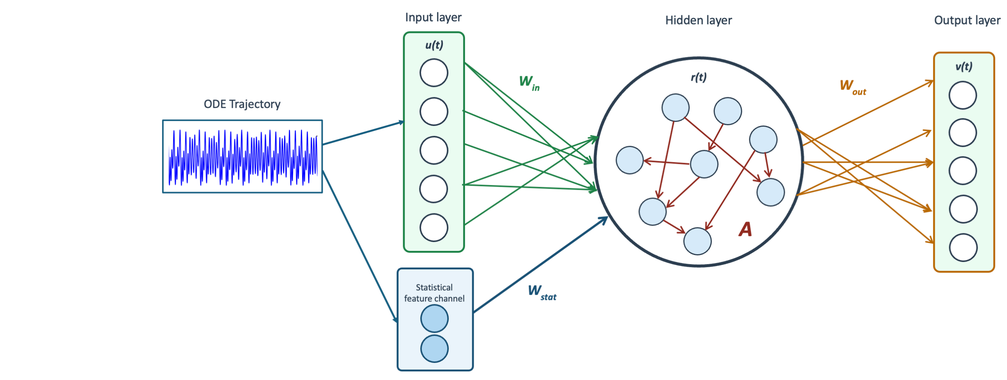
Introduction

- **Problem.** Seizures strike without warning, yet a patient's seizure states are rarely recorded and differ from event to event, and the tissue excitability that drives the transition cannot be measured. A predictor can hardly learn the seizure state itself.
- **Approach.** We learn **normal dynamics only**: a Statistically Conditioned Digital Twin (SCDT) is conditioned on statistics measured from the observed signal and continued beyond its training range. Early-warning indicators only tell that risk is rising; here statistics are **coordinates**, and the rate at which the twin keeps normal dynamics reconstructs where normal dynamics is lost (boundary $\hat{s}_c$). Monitoring would then read the distance between the measured s(t) and $\hat{s}_c$ as risk, not a seizure time.
- **Validation.** The Epileptor model (single node and a three-node network) serves as ground truth (GT), whose transition is known.

Research question Can a digital twin trained only on normal, observable data locate the seizure transition — in statistical coordinates — without ever seeing a seizure?

Method

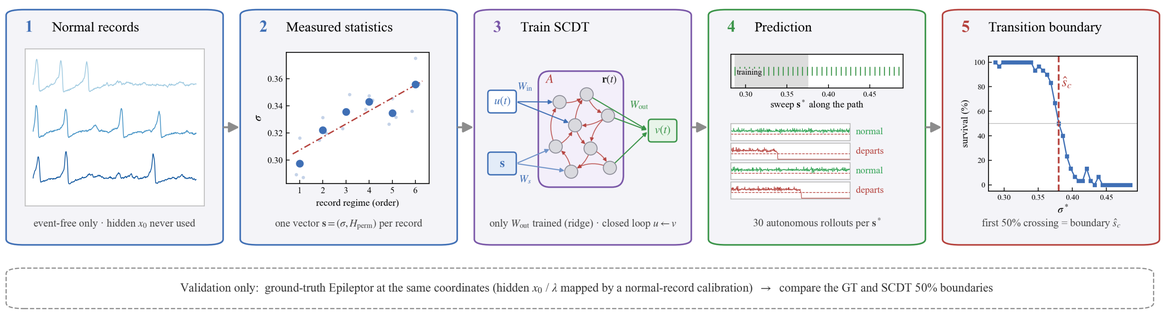
Statistically Conditioned Digital Twin (SCDT)



$$r(t+\Delta t) = (1-\alpha)\, r(t) + \alpha \tanh[\, A\, r(t) + W_{in}\, u(t) + W_s\, \tilde{s}\,], \qquad v(t) = W_{out}\, [\varphi(r(t)), u(t), 1]$$

The SCDT is an Echo State Network with two inputs: the observed signal $u(t) = x_2 - x_1$ and, through a separate static channel, **statistics s measured from the same record** (z-scored, held fixed per record). The statistics act as the coordinates of the dynamical regime, in place of the unobservable excitability. Only the readout W_{out} is trained (ridge regression, one-step prediction); in closed loop ($u \leftarrow v$) the reservoir generates trajectories at any statistical coordinate s*.

Method pipeline



Train

- **Normal regimes only.** The twin is trained on event-free records from several normal regimes; records with any departure are discarded, so it never sees a seizure, and the hidden excitability is never supplied.
- **Noise from residuals.** The one-step prediction residuals of the trained readout form a noise pool; during autonomous generation, residuals drawn from this pool are added at each step, so the twin reproduces the stochastic variability of the real signal.

Statistical conditions — one value per record

- **Single node:** standard deviation σ and permutation entropy.
- **Three-node network:** among eight candidates, a pre-registered λ-free rule (largest between/within-regime variance ratio F) picks **mean** (F = 12.2), then σ.

Prediction and transition boundary

1 Sweep s* along a line through the training statistics, up to two range-widths beyond the training range.

2 Rollouts: 30 stochastic autonomous trajectories per coordinate.

3 Departure: $x_2 - x_1 \le -0.5$ after 5 s (any node), identical for GT and SCDT.

4 Boundary: the first 50% crossing $\hat{s}_c$ of the survival rate S(s*). Control: statistics channel set to zero.

Ground truth: Epileptor model

Six-variable Epileptor (Proix et al., 2017) with additive noise on x_2, y_2:

$$\dot{x}_1 = y_1 - f_1(x_1, x_2, z) - z + I_1, \qquad \dot{y}_1 = 1 - 5x_1^2 - y_1$$

$$\dot{z} = [4(x_1 - x_0) - z - G\, \Sigma_j\, K_{ij}\, (x_{1,j} - x_{1,i})] \,/\, \tau_0$$

$$\dot{x}_2 = -y_2 + x_2 - x_2^3 + I_2 + 0.002g - 0.3(z - 3.5), \quad \dot{y}_2 = (-y_2 + f_2(x_2)) \,/\, \tau_2, \quad \dot{g} = x_1 - \gamma g$$

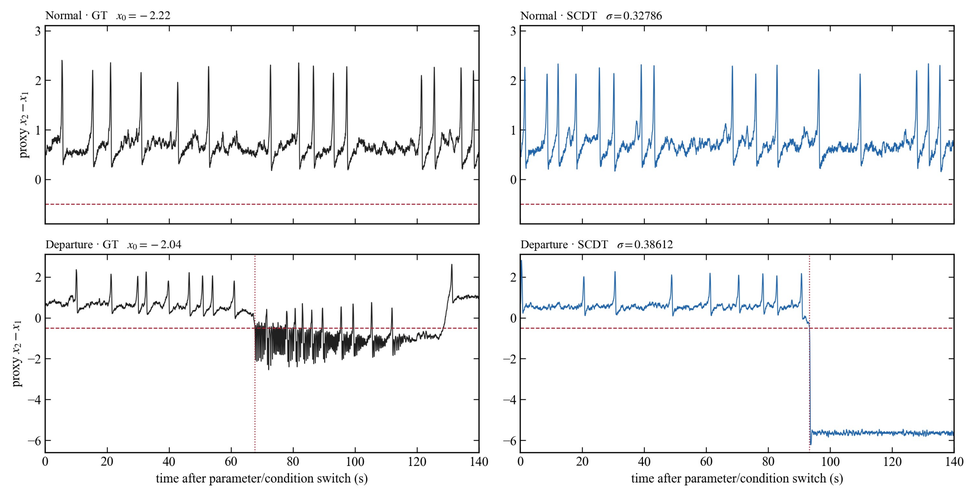
Fast discharges (x_1, y_1), spike-and-wave (x_2, y_2), slow permittivity z. The twin sees only the LFP proxy $x_2 - x_1$ (32 Hz); the excitability is the hidden bifurcation parameter.

- **Single node:** no coupling (K = 0); hidden parameter x_0.
- **Three-node network:** triangle connectome $K_{12} = 1$, $K_{13} = 0.45$, $K_{23} = 0.70$, G = 0.7; one hidden drift λ with $x_{0,i} = \lambda + (0, -0.05, -0.10)$, so node 0 is the most excitable.

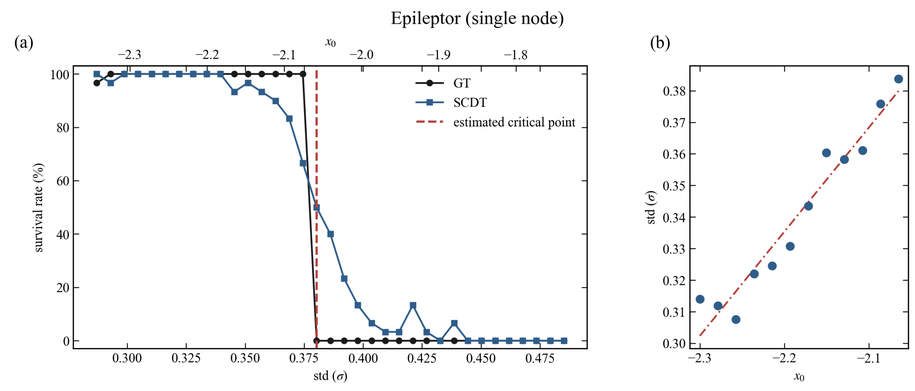
Results

Single-node Epileptor

Conditions σ and permutation entropy; 18 normal records. Beyond the boundary the twin loses normal dynamics, never having seen a departure.



GT (black) and SCDT (blue) trajectories: normal (top) and past the boundary (bottom).

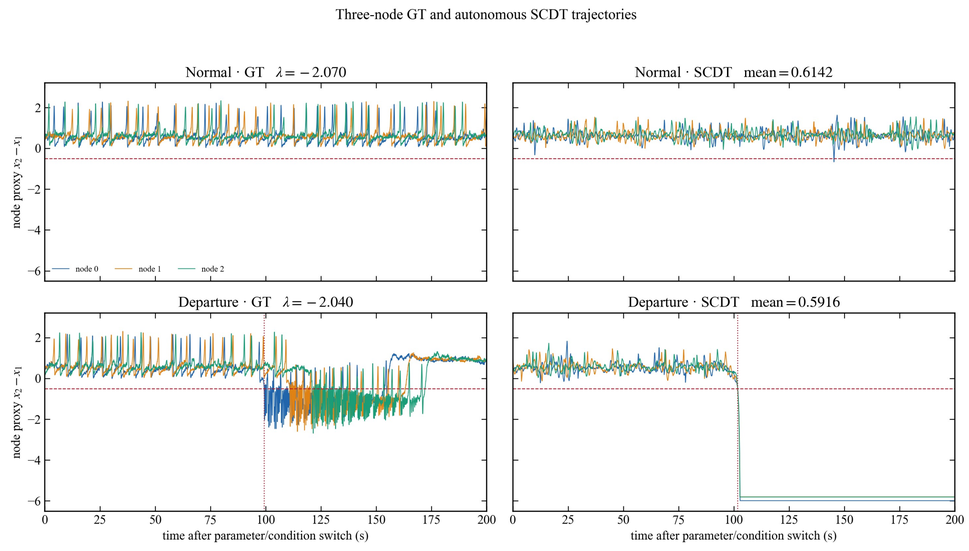


(a) Survival rate along σ, GT and SCDT on the same 35 coordinates. (b) σ vs x_0.

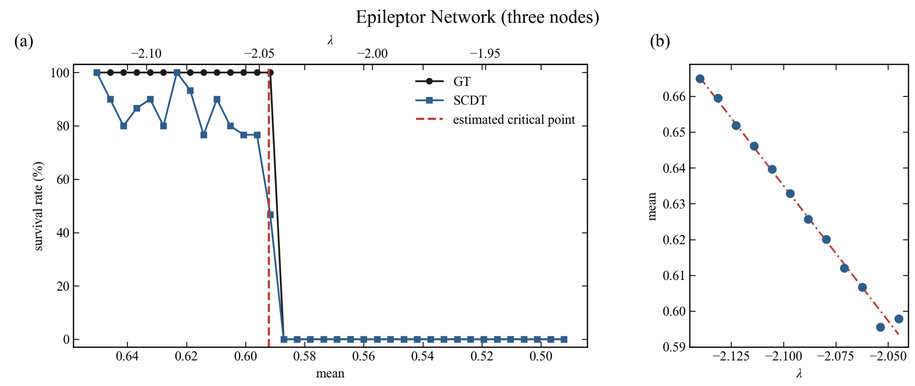
GT boundary	$x_0 = -2.0615 \rightarrow \sigma = 0.3792$
SCDT boundary	σ = 0.3803 (95% CI 0.3745–0.3868)
Difference	+0.0011 σ (0.3%), outside the training range
Control (no statistics)	survival 0.93–1.00 everywhere — no transition

Three-node Epileptor network

52 normal records from 13 regimes. A pre-registered λ-free rule selected the node-averaged **mean** (F = 12.2) as the leading condition.



Three-node GT and SCDT trajectories: normal (top) and past the boundary (bottom).



(a) Survival rate along the mean, GT and SCDT on the same 36 coordinates. (b) mean vs λ.

GT boundary	$\lambda = -2.0438 \rightarrow$ mean = 0.5906
SCDT boundary	mean = 0.5921 (95% CI 0.5907–0.5939)
Difference	+0.0016 (0.27%; Δλ = −0.002, SCDT warns slightly earlier)
Control (no statistics)	≥ 24/30 survive at all 36 points — no transition

Conclusion

- **Transition boundaries from normal data alone:** a twin trained only on normal dynamics located where the normal attractor is lost, within 0.3% of the GT.
- **Limits:** assumes statistics track the transition beyond the training range; intervals cover rollout noise only.
- **Next:** test on human iEEG (SWEC-ETHZ) whether $\hat{s}_c$ can serve as a seizure threshold.

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